PulseIQ — Phase 1 Review & Verification Report

Foundation Phase (Days 1 to 5): Automated ETL, Kimball Star Schema, DQ Gatekeeper, Governed Metrics & Streamlit

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GitHub Repository: github.com/vishesh1603/PulseIQ-ETL	**Git Tag:** phase-1	**Collaborator Added:** fobesitech

1. Phase 1 Foundation: Five-Task Deliverables Table

Task	Objective & Curriculum Spec	Status	Verified Deliverable & Evidence (Repository Links)
Task 1 ETL Pipeline	Single-command pipeline; Pandera schema contracts; quarantine bad rows; row reconciliation (in = passed + quarantined).	**DONE**	src/pulseiq_etl/ (pipeline.py, validate.py). Pandera contracts enforced. Rejected rows quarantined in data/rejected/. Full audit log in logs/last_run.json.
Task 2 Star Schema	Kimball star schema in PostgreSQL 16; SCD Type 2 employee history; zero orphans; diagram committed.	**DONE**	sql/001_schema.sql, sql/004_additional_tables.sql in schema dw. dim_employee has 1,606 versions across 745 staff with btree_gist exclusion constraint. Diagram: docs/warehouse_schema.svg.
Task 3 DQ Gatekeeper	14 automated DQ checks; pass/fail report; reconciliation against all planted issues in dataset.	**DONE**	scripts/run_all_14_dq_checks.py and scripts/compare_dq_known_issues.py. 100% of all 11 planted defects in dq_known_issues caught and documented.
Task 4 Metric Layer	28 official metric SQL views; metric_catalog view; k-anonymity privacy rule (<5/10 suppressed); 5 hand checks.	**DONE**	sql/005_metric_views.sql (views dw.metric_m01_... to dw.metric_m28_... and catalog). All 5 official hand-checks match answer key with 100% precision.
Task 5 Streamlit App	Multi-page Streamlit app; reads exclusively from metric views; finding-first titles; max 3 colours; filters.	**DONE**	app.py and pages/1_Attendance_and_Leave.py. Dockerized service running at port 8501. Uses Okabe-Ito color-blind safe palette (#0072B2, #E69F00, #009E73).

2. Proof It Works: Metric Reconciliation & DQ Check Results

Five Official Metric Hand Checks (Exact Match to Reconciliation Answer Key):

- **M01 Headcount (Sep 2026): 441** employees active on month-end (matches fact_headcount_monthly).
- **M07 Gross Pay (Apr 2025): ■52,740,110.43** unsuppressed (Executive Office ■479,009.00 suppressed at dept grain per privacy rule).
- **M09 Employer Payroll Cost (Apr 2025): ■53,363,392.08** across 386 paid employees.
- **M13 Attendance Rate (Apr 2025, Engineering D01): 0.9509 (95.09%)** (matches reference_summary_dept_month).
- **M16 Company Revenue (Apr 2025): ■71,075,300.00** (ledger accounts 4001–4004).

Check ID	Rule & Description	Target Table	Status	Gatekeeper Action & Finding
DQ-C01	Payroll equals ledger (5001-5005)	fact_payroll vs ledger	**FLAGGED**	Caught June 2026 discrepancy (JE-002094, ■18,450 overage).
DQ-C02	Ledger balances per journal entry	fact_ledger	**PASS**	All 2,524 journal entries balance to the paisa (debits == credits).
DQ-C03	Unique payroll row per person/month	fact_payroll	**PASS**	0 duplicate payroll rows; PK (employee_id, date_key) enforced.
DQ-C04	No orphan employee references	fact_payroll	**PASS**	Zero orphan keys; all employee references resolve to dim_employee.
DQ-C05	No attendance after exit date	fact_attendance	**PASS**	4 planted post-exit records quarantined at ETL ingestion.
DQ-C07	Invoice paid <= total, balance >= 0	fact_invoice	**FLAGGED**	Flagged INV-00385 (overpaid by ■100; negative receivable).
DQ-C08	Expense total = amount + GST	fact_expense	**FLAGGED**	Flagged EXP-00371 (total ■500 above subtotal + GST).

DQ-C09	Sensible task completion dates	fact_task	**FLAGGED**	Flagged 12 negative-cycle tasks & 8 Done tasks without dates.
DQ-C12	SCD Type 2 history non-overlapping	dim_employee	**PASS**	PostgreSQL btree_gist exclusion constraint guarantees no temporal overlaps.
DQ-C14	Small group privacy suppression	Governed views	**PASS**	Enforced in SQL via CASE WHEN count < min_group_size THEN NULL.

3. Mini Challenge: Defining & Proving Metric M29 (WFH Share)

Challenge Prompt: Without notes, define one new metric, write its SQL view, display it on the dashboard, and prove it by hand in under 45 minutes.

• **Metric ID:** M29 | **Metric Name:** Work From Home (WFH) Share (%)
• **Business Question:** What proportion of present employee-days are worked remotely from home?
• **Formula:** (SUM(wfh_days) / NULLIF(SUM(present_days), 0)) * 100 | **Grain:** Department x Month | **Unit:** % | **Privacy Threshold:** 5 employees
• **View Definition:** Implemented in sql/005_metric_views.sql as dw.metric_m29_wfh_share.
• **Dashboard Integration:** Added as a dedicated KPI card on the Streamlit Attendance Page (pages/1_Attendance_and_Leave.py).
• **Hand Check Proof (September 2026):**
  - Total WFH days company-wide = **3,556** | Total present days = **8,655**
  - Calculated WFH Share = 3,556 / 8,655 * 100 = **41.09%**
  - Top Department: Data & AI (D03) at **53.72%** (541 / 1,007), Engineering (D01) at **53.38%** (1,762 / 3,301).
  - Exact match confirmed between Python raw calculation and SQL view output.

4. Self-Check: Data Lineage & Star Schema Concept

Q1: Can I trace any number on my dashboard back to a raw row?
Yes. Every metric is fully traceable through explicit surrogate and business keys. For example, on the September 2026 dashboard, employee EMP-0002 shows 21 present days and 18 WFH days in Finance (D08). In the raw dataset (data/raw/PulseIQ_Star_Schema_Dataset.xlsx, sheet fact_attendance_monthly, row 18), record FA-00018 maps to employee_id = 'EMP-0002' and date_key = 20260930. Joining to dim_employee reveals 7 historical versions; version employee_key = 100009 covers valid_from = 2023-07-18 to 9999-12-31, explaining their current departmental assignment.

Q2: Can I explain a star schema to a friend?
Yes: Imagine a bicycle wheel where the hub is a *Fact table* (the numerical events that happen, like a paycheck issued or a day worked) and each spoke points outward to a *Dimension table* (the context of who, when, where, and what role). Instead of storing employee names, department titles, and job bands redundantly in every payroll row, facts store lean surrogate integer keys. This makes complex business questions like *'What was the average cost per employee in Engineering in Q2?'* lightning fast and consistent through simple joins without duplicating descriptive text.

5. What I Learned (In My Own Words)

Building Phase 1 demonstrated that analytics integrity depends entirely on gatekeeping data at the entrance. Implementing Kimball SCD Type 2 with PostgreSQL's btree_gist exclusion constraint showed how temporal logic prevents duplicate headcounts across reorganizations. I learned that business metrics should never be computed ad-hoc inside BI tools; housing all 28 formulas in governed SQL views with k-anonymity privacy rules guarantees that every stakeholder sees identical, legally compliant numbers.

6. Problems Hit and How They Were Solved

Challenge / Bug Encountered	Root Cause Analysis	Engineered Resolution
Integer Overflow on Ledger Ingestion	Excel stored parent_code in chart of accounts with empty float cells; psycopg coerced NaNs to out-of-range integers.	Built a type-sanitizing loader (scripts/load_additional_tables.py) using NumPy NaN checks to safely map missing codes to SQL NULL.
Temporal Overlaps in Employee History	Reassignment dates in raw HR feeds occasionally overlapped by 1–2 days, which would double-count employee versions in monthly joins.	Added PostgreSQL btree_gist extension with an EXCLUDE USING gist (employee_id WITH =, daterange(valid_from, valid_to, '[]') WITH &&) constraint to enforce strict non-overlapping periods.
Privacy Deanonymization in Small Teams	Executive Office (D00) has only 1 active staff member; displaying average pay or attrition directly exposes personal executive salary.	Built dynamic k-anonymity privacy masks in SQL views (CASE WHEN count < 5 THEN NULL ELSE metric END), suppressing sensitive aggregates while preserving company totals.

7. Plan for Phase 2: Core Analytics & Deep Insights (Days 6 to 10)

• **Attrition & Flight Risk Analytics (Task 6):** Join fact_attrition_risk with dim_employee to model tenure vs. flight risk and identify leading departure indicators.

• **Payroll Equity & Compensation Bands (Task 7):** Compute compa-ratios against role salary bands (dim_role) and evaluate internal pay parity across departments.
• **Productivity & Sprint Velocity (Task 8):** Analyze 28k+ engineering tasks across sprints to evaluate on-time delivery vs. bug reopen rates.
• **Financial & Ledger Correlation (Task 9):** Correlate monthly payroll overhead with revenue realization and DSO collection curves.